

**Weekly Topical Worksheet Pack****CANDIDATE****Urwah****SUBJECT****Chemistry — Cambridge International A Level (9701)****TOTAL MARKS (WEEK)****12 marks****HOW TO USE THIS PACK**

One short set of questions per session, matching that day's topic from the schedule.

Answer in the spaces / on the lines provided. Marks for each part are shown in [brackets].

Mark schemes for the whole week are collected at the end of the pack.

These are original questions written to match Cambridge's exact style and command words — not reproductions of real past paper text.

**THIS WEEK AT A GLANCE**

MONDAY · 17 AUG — TOPIC 1.1: PARTICLES IN THE ATOM & ATOMIC RADIUS (8 marks)

TUESDAY · 18 AUG — TOPIC 1.2: ISOTOPES (4 marks)

**MENTORA ACADEMY · EXCELLENCE IN EDUCATION.****TOPICAL PRACTICE — WEEK 1****Physical Chemistry — Topic 1: Atomic Structure**

Monday 17 August 2026 – Friday 21 August 2026

## Question Pack

## MONDAY · 17 AUG — TOPIC 1.1: PARTICLES IN THE ATOM &amp; ATOMIC RADIUS

**1 Multiple Choice** Syllabus 1.1

For each question, put a cross in the box next to your chosen answer.

- (a) Which particle has a relative charge of +1 and is found in the nucleus? [1]  
A electron  
B neutron  
C proton  
D positron
- (b) An atom has proton number 15 and nucleon number 31. How many neutrons does it contain? [1]  
A 15  
B 16  
C 31  
D 46

**2 Sub-Atomic Particles** Syllabus 1.1

Complete the table to show the relative charge and relative mass of each sub-atomic particle.

- (a) State the relative charge and relative mass of a proton, neutron, and electron. [3]

**3 Behaviour in an Electric Field** Syllabus 1.1

A beam containing protons, neutrons and electrons is passed through a uniform electric field created by two charged parallel plates.

- (a) Describe and explain what happens to the protons as they pass through the field. [2]
- (b) Explain why the neutrons are undeflected. [1]

**Question Pack****TUESDAY · 18 AUG — TOPIC 1.2: ISOTOPES****4 Defining & Representing Isotopes** Syllabus 1.2

- (a) Define the term isotope. [2]
- (b) Chlorine exists as two isotopes. Using the correct notation, write the full isotopic symbol for chlorine-35 and chlorine-37. [2]

## Mark Scheme

## MARK SCHEME – WEEK 1

| Q    | MARKING POINTS                                                                                                                                                | MARKS |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1(a) | C (proton has relative charge +1 and is found in the nucleus)                                                                                                 | 1     |
| 1(b) | B ( $31 - 15 = 16$ neutrons)                                                                                                                                  | 1     |
| 2(a) | Proton: charge +1, mass 1; Neutron: charge 0, mass 1; Electron: charge -1, mass $1/1840$ (or negligible). 1 mark per correct row.                             | 3     |
| 3(a) | Protons are deflected towards the negative plate (1); because they have a positive charge and are attracted to the oppositely charged plate (1).              | 2     |
| 3(b) | Neutrons have no charge / are electrically neutral, so there is no electrostatic force acting on them (1).                                                    | 1     |
| 4(a) | Atoms of the same element (same number of protons / same proton number) (1); with different numbers of neutrons (different nucleon number / mass number) (1). | 2     |
| 4(b) | Correct isotopic notation showing mass number and atomic number for both Cl-35 and Cl-37 (1 mark each).                                                       | 2     |